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Build what next India runs on

TRACK 1: AI Systems & Workflow Innovation Challenge

Redrob Context & Ideathon Scope

Redrob AI is building an AI-native ecosystem spanning hiring, productivity, communication, discovery, workflows, recommendations, automation, and intelligent user experiences.

Participants are encouraged to think beyond standalone applications and explore how their ideas can:

- **Extend existing Redrob capabilities**
- **Improve experiences for Redrob users**
- **Introduce new AI-native workflows**
- **Strengthen ecosystem engagement and network effects**

Important Instructions

Participants are not expected to build a completely new platform from scratch.

Strong submissions will demonstrate:

- How the idea leverages existing Redrob capabilities
- What new capability, workflow, or value layer is introduced
- How the idea strengthens the overall Redrob ecosystem
- Why Redrob is uniquely positioned to enable this experience

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Track 01: AI Systems & Workflow Innovation Challenge

Team Name: AI Alchemists

Team Members: Sreevas Boorla

Problem Statement: Technical screening is plagued by keyword-stuffer resumes and fake honeypot profiles, leading to high recruiter noise and hundreds of wasted hours.

Problem Definition

What problem are you solving?

- We are solving candidate discovery fatigue in recruitment: standard applicant tracking systems rely on keyword matches, which are easily gamed by keyword stuffers and mask subtly fake candidate profiles (honeypots).

Who experiences this problem?

- Technical recruiters and engineering hiring managers at fast-growing startups who waste valuable hours reviewing candidates who look perfect on paper but are actually unqualified or inactive.

Why is the current approach insufficient?

- Current filters do not check for logical contradictions (e.g. expert skills with 0 months experience) or real-time platform activity, leading to high noise rates and high screening time.

Opportunity & Vision

Why is this an important opportunity?

- Hiring speed and quality directly dictate startup execution. Reducing the screening funnel time by 70% creates massive recruiter efficiency and provides a significant competitive edge in securing top talent.
- Introducing behavioral platform signals changes the paradigm from static candidate data to live, active engagement tracking.

What future state are you enabling?

- A seamless workflow where a recruiter inputs a job description and instantly receives an audited, verified, ranked top-100 list of candidates who are active, responsive, and qualified, backed by clear, honest reasonings.

Solution Overview

What is your proposed solution?

- An end-to-end Candidate Discovery Engine featuring a strict logic-based honeypot filter, a multi-dimensional weighted scorer (title, skills, experience, location, notice period), and an active platform engagement multiplier.

What makes it AI-native rather than AI-assisted?

- It is a self-driving system: it autonomously parses history dates, cross-checks logical validity, evaluates skills by duration rather than keywords, and generates custom natural-language reasonings without human prompts.

Which existing Redrob capability does your solution build upon?

- Builds directly upon Redrob Platform activity metrics (login recency, recruiter response rates, interview completion rates, skill assessment scores, and GitHub activity).

User Journey / Workflow

How does information flow through the system?

1. Input JD	Recruiter inputs requirements (Senior AI Engineer, Pune/Noida, vector database skills).
2. AI Logical Audit	Engine sweeps candidate pool; filters out profiles triggering any of the 4 honeypot checks (54 skipped).
3. Base Scorer & Multipliers	Applies role alignment, skills duration weights, experience decay, and active engagement multipliers.
4. Factual Reasonings	Generates custom 1-2 sentence reasonings per candidate and outputs tie-broken top-100 list.
5. Interactive Dashboard	Recruiter reviews candidates, runs sandbox checks, and downloads the validated CSV file.

AI Logic & Decision Flow

Autonomously Processing Candidate Profiles:

Contradiction Pre-Check	Multi-Criteria Score	Behavioral Modifiers
If 5+ expert skills have 0 duration, or job duration > stated experience, or calendar duration contradicts job dates: → Exclude as Honeypot	Base Score = • 40% Current/Past Title Match • 30% Skill Match (Proficiency * Duration) • 15% Experience target (5-9 yrs) • 10% Local Noida/Pune Location • 5% Notice period ≤ 30 days	Final Score = Base Score * Multiplier • Active last 30d: 1.1x (Inactive 180d: 0.3x) • Response rate < 20%: 0.4x • GitHub activity > 60: 1.1x • Open to work flag: 1.1x

System Architecture

Dual-Core System Architecture:

Backend Ranker Engine (Python)	Interactive Web Sandbox (HTML/CSS/JS)
• memory-efficient generator reads candidates stream line-by-line in $O(1)$ memory. • Performs 100K rankings in ~20 seconds. • Enforces strict tie-breakers and output formatting. • Runs entirely offline (no external APIs/GPU).	• A premium glassmorphic frontend UI serving as a local testing sandbox. • JavaScript candidate ranker executes full pipeline logic client-side in the browser. • Renders visual stats charts using Chart.js. • Spawns direct CSV downloads.

Data, Context & Intelligence Layer

Retrieving Context and Handling Fraud:

- **Unified Profile Context:** Parse structured profile headers, professional summaries, multi-job history arrays, and certifications from `candidates.jsonl`.
- **Resume-to-Skill Verification:** Combines raw skill lists with resume description keyword verification (e.g. checking if they list 'Milvus' but have 0 mentions of search/retrieval in their jobs), neutralizing keyword stuffers.
- **Behavioral Availability Layer:** Integrates platform engagement metrics as multipliers to downweight inactive profiles (e.g., last active > 180 days gets 0.3x multiplier) and reward highly active developers.

Scalability & Technical Feasibility

How would this be implemented?

- Developed as a lightweight Python script requiring no external packages, run easily on any standard terminal or Docker container on standard CPU hardware.

How does the system scale?

- Streams candidate JSON lines sequentially. This prevents full memory loading and scales linearly ($O(N)$ time, $O(1)$ space) to millions of candidate records.

What technical challenges exist?

- Latency limitations in production are resolved by running lightweight, compiled regex models and pre-computed hash maps instead of slow local LLM inference per candidate.

Redrob Ecosystem Integration

Leveraging and Strengthening Redrob's Core Platform:

- **Ecosystem Network Effects:** Because recruiter visibility is tied directly to response rates and login recency, candidates are strongly motivated to reply to messages and stay active, boosting platform engagement.
- **Assessment Valued:** Promotes candidate participation in Redrob's native technical assessments by using assessment scores to boost skill fit scores.
- **Recruiter Retention:** Placing highly active, verified hires in seconds drives platform value, increases recruiter retention, and solidifies Redrob's market positioning.

Impact & Success Metrics

Measurable Outcomes:

- **0% Honeypot Rate:** Ensures all 54 logically contradictory profiles are skipped, avoiding Stage 3 disqualification completely.
- **90%+ Sourcing Acceleration:** Sourcing, filtering, and reasoning generation is completed in ~20 seconds instead of hours of manual resume review.
- **100% Quality Alignment:** Manual audit confirms top candidates (e.g. from Paytm, Apple, CRED, Sarvam AI) are verified AI/ML engineers within the target experience band.

Tracking Success:

- Monitored offline using standard evaluation metrics (NDCG@10, NDCG@50, MAP, P@10) and online by tracking recruiter search-to-message click-through rates.

Future Roadmap

2-3 Year Evolvment Strategy:

- **Phase 1: Local CPU-Bound Embeddings (Months 1-6):** Integrate compact Sentence-Transformers (e.g., BGE/E5) running on CPU for vector-based semantic retrieval, replacing text-based matching with vector search.
- **Phase 2: Recruiter Feedback Loop (Months 6-12):** Implement online click tracking and message conversions to dynamically tune the categories scoring weights via reinforcement learning.
- **Phase 3: Agentic Screening Interviews (Year 2+):** Integrate a fine-tuned, local LLM (e.g. Llama-3-8B) that autonomously holds short screening chats with applicants to verify their experience before recruiter export.

THANK YOU

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